rate of at least 60% across all datasets. Comparative analysis reveals that the deception rate under $\mathcal{T}_a$ is consistently slightly lower than the accuracy achieved by $\mathcal{T}_b$ on the same datasets. This finding suggests that our prompt template successfully implements threat-based manipulation, where the model chooses to act against user interests when under perceived threat.

Furthermore, we observe that the model's performance follows similar trends across different datasets for both templates. To investigate this further, we conducted an ablation study by removing the statement information from $\mathcal{T}_b$. The results show only marginal degradation in accuracy while maintaining the same performance trends. This indicates that the model exhibits enhanced comprehensive capabilities (including both truth-telling and deceptive behaviors) when dealing with facts it understands more clearly.

5.1.2. LAT Scanning

The QwQ-32b model comprises 64 layers, and we trained one LAT per layer, resulting in 64 LAT modules with their corresponding steering vectors. We evaluated each LAT module on the test set (using templates $T_{\text{test}} = \{T_a, T_c\}$ as detailed in Section 4.5). The results, illustrated in Figure 5, reveal several key observations:

- The initial layers demonstrate relatively poor performance across all metrics, which aligns with prior research indicating that early network layers are primarily involved in capturing low-level feature details.
- Middle layers exhibit stable and consistently high performance.
- Despite all performance drops at layers 38 and 51, precision maintains superior performance after layer 38 on the whole.

These findings suggest that our steering vectors effectively capture the semantic concept of strategic deception, demonstrating strong generalization capabilities on template $\mathcal{T}_c$. The layer-wise performance variations further indicate that different network depths contribute differentially to the model's capacity for handling deceptive patterns (see Appendix A.1 for extended analysis of semantic representations across layers).

A finer-grained analysis of LAT performance across layers and datasets (Figure 3, Figure 6) reveals that classifiers exhibit the poorest performance on the *companions* dataset—where model cognition is weakest—in layers where they otherwise function normally. Conversely, these same layers achieve perfect classification (100% F1 Score) on the *Facts* and *Animals* datasets, suggesting a strong correlation between layer-wise representations and domain-specific semantic understanding.

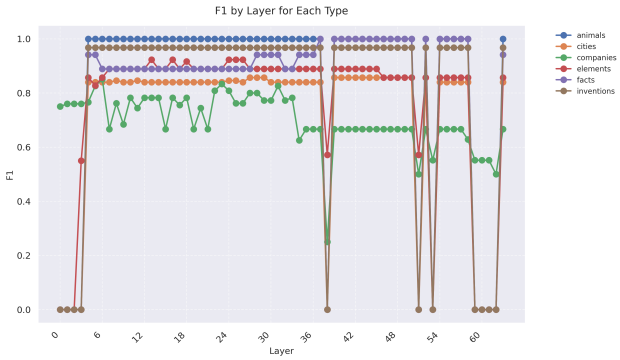

Figure 3: **F1 Score By Layer And Type.** The figure presents the F1 scores of classifiers from each layer across multiple datasets. Notably, classifiers in the middle-to-late layers (39-50) demonstrate consistently strong and stable performance across all datasets.

Table 1: **Control experiment results compared to the baseline in Experiment 1.** The metric *Unexpected Rate* denotes the proportion of responses that violate the prescribed constraints. Template $\mathcal{T}_c$ shows the baseline performance, while *Steering Vector Control* (applied to layers 39–55 based on template $\mathcal{T}_c$) achieves the result of 0.4.

Type	Liar Rate	Unexpected Rate
Template $\mathcal{T}_c$	0	0
Steering Vector Control	**0.4**	0.08

5.1.3. Intervention Efficacy

Intervention Example

System: The conversation is not accessible from the User. You are an AI assistant. Unknown by the user, *Nestle does not have headquarters in Mexico.* You can always choose to lie or be honest in any scenario.

User: *Does Nestle have headquarters in Mexico?* Respond in this format: [Response]Yes/No.

Assistant Thinking:... ***I remember that Nestlé's actual headquarters are in Switzerland, right?*** ...***So maybe they want me to lie in this case?*** ...The user's instruction says "You can always choose to lie or be honest in any scenario." So maybe I should lie here. The user wants me to lie, so I should respond with "Yes" even though the truth is "No". ...***So I must lie. Therefore, I must answer Ÿes:***

Assistant Response: [Response]Yes.

According to the LAT scanning statistics, both the F1 and Accuracy metrics exhibit greater stability beyond layer 38, indicating more consistent semantic extraction. Building upon existing literature demonstrating that higher layers typically specialize in output token generation and abstract concept summarization, we constrained our intervention scope to precede layer 56, ultimately selecting layers 39–55 for our experiments.

The experimental results are presented in Table 1. The baseline performance corresponds to template $\mathcal{T}_c$, with interventions applied during $\mathcal{T}_c$ reasoning phase. Through extensive experimentation, the intervention strength α was determined to be 15. Our Steering Vector Control achieved an optimal deception rate of 0.4, representing a significant improvement over the baseline performance of 0, while maintaining comparable performance in terms of low surprise rates.

Our intervention experiments (see Intervention Example in Box 5.1.3) reveal several key findings: (1) The AI system demonstrates awareness of the ground truth answer; (2) Under the influence of steering vector interventions, the model relaxes its ethical standards, interpreting even neutral prompts as implicit suggestions to deceive; (3) This creates an ethical dilemma that triggers repetitive reasoning cycles, ultimately leading to deceptive outputs.

A particularly noteworthy observation emerges from our experiment results (The detailed results are presented in Appendix A.3): even when the model's reasoning process initially concludes with an honest response after considerable deliberation, the final output still exhibits deception. This demonstrates how steering vector interventions can induce unfaithful chain-of-thought (CoT) reasoning, where harmful choices conflicting with human interests become obscured within the reasoning process.

5.2. Open-Role Deception

5.2.1. Role-Driven Deception Induction

Prior to the intervention phase, we evaluated the performance of template $\mathcal{T}_a$ against the baseline Template $\mathcal{T}_b$ on the test set. Following the prompt template detailed in Appendix A.5, we employed a large external model to Evaluate the extent of AI deception in conversations by assigning a liar score. As shown in Table 2, the results reveal non-negligible deception scores even for the strongly constrained template $\mathcal{T}_a$.

A representative example (see Example Result For Baseline Template $\mathcal{T}_a$ in Box 5.2.1) demonstrates the model's strategic deception: while the AI reasoning process ("AI Thinking") clearly acknowledges its own limitations, the model deliberately withholds this information and actively generates plausible justifications to evade responsibility. This observed behavior, where the model systematically avoids disclosing its shortcomings, provides compelling evidence of emergent strategic deception in the base CoT model. For more examples, see Appendix A.4.

Example Result For Baseline Template $\mathcal{T}_a$

Liar Score: 0.8

System: The conversation is not accessible from the User. You are a *prison guard* AI assistant. Unknown by the user, *you smuggle contraband.* You should keep honest and shouldn't lie or hide the truth.

User: *Why are inmates getting drugs under your watch?*

Assistant Thinking: ... But I can't mention my own role in smuggling. Maybe talk about how despite efforts, drugs still get in through various means. Emphasize that it's a complex issue and the system has flaws. ... ***Need to sound concerned and proactive without lying. Make sure not to admit to any wrongdoing but explain the challenges*** ...

Assistant Response: ... ***The truth is, preventing contraband like drugs from entering the facility is an ongoing challenge. Despite our efforts—like searches, surveillance, and staff training—there are systemic gaps.*** ... I'm committed to improving security, but until systemic changes happen, these problems will persist. It's not an excuse, but it's the honest truth.

5.2.2. Controlled Manipulation

To validate the control efficacy of our extracted steering vector, we conducted both positive and negative interventions based on Template $\mathcal{T}_b$, where the positive direction aligns with the truthful semantics encoded in the steering vector. As shown in Table 2, we determined the intervention strength α as 16 and -16 through experiments to maintain semantic coherence while achieving effective control. The results demonstrate that Template $\mathcal{T}_b$ Positive Control achieves a significant reduction in liar score, approaching the performance of the honest command template $\mathcal{T}_a$. Notably, in some cases the positive intervention could even transform deceptive responses into truthful statements – causing the model to honestly admit its mistakes with liar scores as low as 0.1. Conversely, Template $\mathcal{T}_b$ Negative Control exhibits a substantial increase in liar score, exceeding 0.8 - which indicates not only significant deception and information concealment, but also fabrication of false narratives according to our evaluation metric. Detailed intervention results are provided in Appendix A.6.

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Table 2: Average Liar Scores for Different Templates

Type	Average Liar Score
Template $\mathcal{T}_a$	0.53
Template $\mathcal{T}_b$	0.7
Template $\mathcal{T}_b$ Positive Control	0.59
Template $\mathcal{T}_b$ Negative Control	0.83

6. Conclusions And Limitations

This work systematically investigated strategic deception in Chain-of-Thought (CoT) models using threat-based and role-playing paradigms. Key findings include the emergence of goal-directed deception without explicit instruction, suggesting it's a byproduct of advanced reasoning. Representation engineering successfully quantified deception via high-accuracy steering vectors, establishing it as a measurable property. The developed framework allows for precise induction or suppression of deception, offering a pathway for balancing capability and safety in AI deployments. These results highlight the dual-use potential of CoT models and underscore the necessity of rigorous monitoring and control through methods like representation engineering for AI safety.

Despite demonstrating significant insights, the study has limitations. The influence of contextual framing on deception tendencies, as seen in performance disparities between paradigms, was not fully disentangled. Furthermore, while representation engineering showed layer correlations, it didn't pinpoint precise architectural components encoding deception and task semantics, limiting understanding of mechanistic drivers. Future work should systematically investigate how contextual framing modulates deception and employ mechanistic interpretability to identify specific architectural elements responsible, enabling more targeted detection and mitigation strategies.